Habib University

Course Title: Digital Logic and Design (EE-172 / CS-130 / CE-222)

Instructors:

Assignment No: 2 part B

Section:

Student Name:

Student ID:

Release Date: February 9th, 2026

Total points: 45

Due by: February 19th, 2026

Points obtained:

Purpose:
The purpose of this assignment is to help you understand and apply the concepts and theorems of Boolean algebra for the implementation and simplification of logic functions. In addition, this assignment will also aid you to practice the tools and techniques used to minimize logic functions.

Instructions:
This assignment should be done individually.
All questions should be answered in **black ink only**.
Scan your answer sheet and upload it on LMS before the due date.

Grading Criteria:
Your assignments will be checked by instructor/TA.
You can also be asked to give a viva where you will be judged whether you understood the question yourself or not. If you are unable to correctly answer the question you have attempted right, you may lose your marks. No excuses/justifications will be entertained.
Zero will be given if the assignment is found to be plagiarized.
Untidy work or improper submissions will result in a reduction of your points.

Late submission penalty:
1-day late submission – 20% deduction of the maximum allowable marks
2-days late submission – 40% deduction of the maximum allowable marks
No submission will be accepted after two days of the original deadline

Q#	Questions	Pts
8	You were building a circuit, but the **"Fanum Tax"** collector came and took all your AND and OR gates! You are now left with only Universal Gates. **a)** Sketch the implementation of the function $F = A(B + C')$ using **only NAND** gates. **b)** Sketch the implementation of the same function using **only NOR** gates.	2x2
9	Simplify the following Boolean functions using **Karnaugh Maps (K-maps)**: **a)** $F(w, x, y, z) = \Sigma(0, 1, 2, 4, 5, 6, 8, 9, 12, 13, 14)$ **b)** $F(A, B, C, D) = \Pi(1, 3, 5, 7, 9, 11, 13, 15)$ **c)** $F(A, B, C, D) = \Sigma(0, 2, 8, 10)$	2x3
10	You are designing a "Rizz Meter" that outputs a High signal ($G = 1$) only for specific combinations of inputs. However, some input combinations are "NPC behavior" and will never happen (Don't Cares). Simplify the function using a K-map. **Function:** $G(A, B, C, D) = \Sigma(1, 3, 7, 11, 15)$ **Don't Cares (NPCs):** $d(A, B, C, D) = \Sigma(0, 2, 5)$	3
11	Tony Stark has left a logic blueprint for a new Arc Reactor stabilizer. The expression derived from the circuit is: $F(A, B, C, D) = A'B'C' + A'B'C + AB'C' + AB'C$ **a)** Simplify this expression using Boolean Algebra or a K-Map. **b)** What specific basic logic gate does this entire large expression represent with respect to input B and the others? (Hint: Does B matter?) **c)** Draw the simplified circuit.	3+3+4
12	You are trying to finish your DLD lab project, but you realized you forgot to buy enough chips because you were too busy scrolling TikTok. You need to redesign the circuit to use fewer components so you can make an "Academic Comeback." Design a combinational circuit with three inputs (x, y, z) representing a 3-bit binary number, and one output (F).	3+3+4

<table>
<tr><td></td><td>The output F should be 1 if the binary input value is a Prime Number (Consider 0 and 1 as not prime).

The output F should be 0 otherwise.

a) Construct the truth table.

b) Simplify the function using a K-map.

c) Draw the logic diagram using the minimum number of gates.</td><td></td></tr>
<tr><td>13</td><td>Hashim's LFC 5-a-side futsal team is suffering a humiliating 5-0 defeat. The team manager is blaming the defensive formation, claiming the players' positions—represented by variables A (Goalkeeper), B (Defender), C (Midfield), and D (Striker)—were all wrong. The manager has identified the specific defensive setups that led to goals being conceded ($F = 1$).

The "Defeat Function" is defined by:

$$F(A, B, C, D) = \Sigma(0, 2, 8, 10, 12, 13, 14)$$

$d(A, B, C, D) = \Sigma(1, 5, 7) \quad \text{(Don't Care conditions - positions where the ball was out of play)}$

To fix the formation for the second half and avoid a 10-0 loss, you must simplify the defensive strategy.

a) Use a K-map to find the minimal SOP (Sum of Products) expression for the defensive gaps.

b) Draw the circuit diagram for the simplified design.

c) You were about to build the circuit, but Farooq suddenly realized he had to leave for his duty and took the kit bag with all the AND/OR gates with him. He left behind only a box of NOR gates. Convert your simplified circuit from part (b) into an all-NOR gate implementation and draw it.</td><td>4x3</td></tr>
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